MOHAMMAD ROHAAN

BSCS Candidate · FAST NUCES, Islamabad
Software Engineering · AI/ML · Robotics & Embedded Systems

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OBJECTIVE

BSCS candidate at **FAST NUCES**, **Islamabad** (expected graduation Jan 2027). Delivered end-to-end projects in **AI/ML (NLP quiz generation, YOLOv8 object detection, tabular ML)**, software engineering (**Spring Boot** library platform with Scrum and full test coverage), database systems (SQL Server + PHP hospital application), and systems programming (OpenMP K-means analysis, SpMV/HPCG optimization, **Git-style version control** in C++). Seeking software engineering, **AI/ML, or embedded robotics internships** where I can ship reliable, measurable work.

EDUCATION

FAST NUCES — Islamabad, Pakistan
Bachelor of Science in Computer Science (BSCS) Expected Jan 2027 Final-year BSCS candidate

Relevant Coursework

Data Structures & Algorithms, Operating Systems, Computer Networks, Database Systems, Software Engineering, Probability & Statistics, Parallel & Distributed Computing, Numerical Computing / HPC, Design & Analysis of Algorithms, Embedded Systems, Machine Learning / Data Science (project-based), Computer Vision (project-based)

TECHNICAL SKILLS

Languages: Python, Java, C, C++, C#, JavaScript, SQL, PHP, Assembly x86 (basics)
AI / ML / Data: Scikit-learn, pandas, NLP (TF-IDF), CNN / deep learning (basics), YOLOv8,Streamlit, model evaluation , statistics & distribution analysis, data preprocessing, feature engineering
Software Eng: Java 17, Spring Boot 3.5, Spring MVC, Thymeleaf, Spring Security, Spring Data JPA, Hibernate, MySQL 8, Scrum / Agile, REST APIs, Git, Maven
Web & Databases: HTML, CSS, JavaScript, PHP, SQL Server, T-SQL, ER / EER modeling, normalization
Systems: OpenMP, parallel & distributed computing (basics), multithreading, mutexes / semaphores, benchmarking, DSA, numerical computing (SpMV / HPC)
Embedded / Robotics: Arduino, ESP32, Arduino IDE, Raspberry Pi, ultrasonic & common sensors, MATLAB (robotics simulation & kinematics basics), control themes (course/projects)
Tools & Other: Linux, shell scripting, Cisco Packet Tracer (course labs), cybersecurity fundamentals

SELECTED PROJECTS

AI Quiz Generator (Python, Streamlit, scikit-learn) — RACE-based quiz system with dual ML pipelines
Shuttlecock Detection (YOLOv8, transfer learning) — 15k-image fine-tune for robotic pickup
K-means Triangle Inequality (OpenMP) — Elkan vs Lloyd, scheduling & dimension benchmarks.
LibraryMS (Spring Boot 3.5, Java 17, MySQL, Scrum) — Role-based library platform.
Robot Energy Prediction (Python, scikit-learn) — Energy modeling from operational features.
Fraud Detection (Python, scikit-learn) — Supervised fraud classification on imbalanced data.
Hospital DB System (SQL Server, PHP, ER/EER) — Normalized schema, 12 SQL queries, PHP/JS.
Multi-Agent Game AI (Python) — Adversarial search / game-playing agents with board I/O.
Git Lite (C++) — Mini Git: commit DAG, hashing, diff/merge; DSA/systems fundamentals.
SpMV / HPCG Optimization (C++) — Sparse kernel performance: correctness, benchmarking.